Cef-4®

For Intravenous or Intramuscular Use

COMPOSITION :-
Each vial contains :-
Sterile Ceftazidime pentahydrate (buffered) equivalent to Ceftazidime 500 mg, 1 g.

PRODUCT DESCRIPTION :-
White to slightly yellow sterile powder or crystalline powder.

PHARMACOLOGICAL INFORMATION :-
Cef-4® is a broad-spectrum antibiotic, with enhanced activity against *Pseudomonas aeruginosa*. Mechanism of action: Inhibits bacterial cell wall synthesis by binding to one or more of the penicillin-binding proteins (PBPs) which in turn inhibit the final transpeptidation step of peptidoglycan synthesis in bacterial cell walls, thus inhibiting cell wall biosynthesis. Bacteria eventually lyse due to ongoing activity of cell wall autolytic enzymes (autolysins and murecin hydrolases) while cell wall assembly is arrested. It is highly stable to hydrolysis by most β-lactamases produced by gram-negative and gram-positive bacteria and consequently is active against *S.aureus*, *Steptococcus pneumoniae*, *β-hemolytic streptococci*, *E.coli*, *H.influenzae*, *Klebsiella* sp., *Neisseria meningitidis*, *Proteus mirabilis*, *Proteus vulgaris*, *Enterobacter* sp., *Citrobacter* sp., *P.aeruginosa*, *Serratia* sp.

PHARMACODYNAMICS/KINETICS :-
Cef-4® administered by the parenteral route reaches high and prolonged serum levels. Therapeutically effective concentrations are still found in the serum 8-12 hours after IV and IM administration.
Distribution :- Widely distributes throughout the body including bone, bile, skin, CSF (diffuses into CSF with higher concentrations when the meninges are inflamed), endometrium, heart, pleural and lymphatic-fluids.
Half-Life :- 1-2 hours (prolonged with renal impairment). For Neonates < 23 days, half-life is 2.2-4.7 hours.
Time to peak serum concentration :- Following intramuscular (IM) administration of 500 mg and 1 g doses of Ceftazidime to normal adult volunteers, time to peak serum concentration was approximately 1 hour.
Protein binding : - Approximately 10%
Elimination :- By glomerular filtration with 80% to 90% of an IM or IV dose of ceftazidime excreted as unchanged drug within 24 hours.

INDICATIONS :-
Cef-4® is used in the treatment of susceptible infections including :-
- Lower Respiratory Tract Infections
- Urinary Tract Infections
- Skin and Soft Tissue Structure Infections
- Bacterial Septicemia
- Bone and Joint infections
- Gynecological infections, including endometritis, pelvic cellulitis and other infections of the female genital tract
- Intraabdominal infections
- CNS infections
- Severe and life-threatening infections

CONTRAINDICATIONS :-
Cef-4® is contraindicated in patients who have shown hypersensitivity to ceftazidime or the cephalosporin group of antibiotics.

SIDE EFFECT/ADVERSE REACTIONS :-
Ceftazidime is generally well tolerated. The incidence of adverse reactions associated with the administration of ceftazidime was low. The most common were local reactions following IV injection and allergic and gastrointestinal reactions. Other adverse reactions were encountered infrequently. No disulfiram-like reactions were reported.
Gastrointestinal :- Diarrhea (1.3%)
Local :- Pain at injection site (1.4%)
Miscellaneous :-
2% : Hypersensitivity reaction.
< 1% : Anaphylaxis, fever, CNS (headache, dizziness, paresthesia, bad taste, pruritus, neuromuscular excitability, rash, genitourinary (candidiasis, vaginitis), hemolytic anemia, phlebitis.
Laboratory test change :- Transient changes noted during Ceftazidime Therapy include : eosinophilia, positive Coombs' test without hemolysis, thrombocytosis, and slight elevations in one or more of the hepatic enzymes, ALT (SGPT), AST (SGOT), LDH, GGT and alkaline phosphatase.

WARNINGS/PRECAUTIONS :-
Use with caution in patients with a history of hypersensitivity reactions to ceftazidime, cephalosporins and penicillins especially IgE-mediated reactions (e.g. anaphylaxis, urticaria). If an allergic reaction to ceftazidime occurs, discontinue the drug. High dosage of cephalosporin antibiotics should be given with caution to patients receiving concurrent treatment with nephrotoxic drugs, eg. aminoglycosides or potent diuretics (e.g. furosemide). The combinations are suspected of affecting renal function adversely. It is necessary to reduce the dosage of antibiotic eliminated via the kidneys according to the degree of reduction in renal function Treatment with antibacterial agents alters the normal flora of colon and may cause antibiotic-associated colitis or colitis secondary to *C.difficile*.

USE IN PREGNANCY & LACTATION :-
Pregnancy Risk Factor B : No experimental evidence of embryopathic or teratogenic effects attributes to ceftazidime, but it should be administered with caution during the first trimester and in early infancy. Cef-4® is excreted in human milk in low concentration, so caution is exercised when Cef-4® is administered to nursing mother.

RECOMMENDED DOSAGE :-
Determine dosage and route by the susceptibility of the causative severity of infective and patient's condition and renal function.
Adults : The usual dose is 0.5-2 g every 8-12 hours.
In severe infections, the dosage may be increased up to 9 g daily.
Infants and Children : Children > 2 months is 30-100 mg/ kg/ day in divided doses every 8 or 12 hours. The dosage of children between 2 months and 1 year is generally 25-50 mg/ kg twice daily.
Dose up to 50 mg/ kg/ day 3 times daily, to a maximum of 6 g daily, may be given to infected immunocompromised or fibrocystic children or to children with meningitis.
Neonates and children up to 2 months : A dose of 25-30 mg/ kg/ day in divided doses every 12 hours has proved to be effective.
Patients with renal function impairment
Give an initial loading dose of 1 g then the dosage may need to be reduced as follows:

Creatinine clearance (mL/min)	Recommended dose (g)	Frequency of dosing (hr)
31-50	1	12
16-30	1	24
6-15	0.5	24
≤ 5	0.5	48
Hemodialysis patients	1	after each hemodialysis
Peritonial dialysis patients	0.5	period 24

Preparation of solution

Vial size	Amount of Diluent to be Added (mL)	Approximate Concentration (mg/mL)
500 mg IM	1.5	280
500 mg IV	5	100
1 g IM	3	280
1 g IV	10	100

Actual Size 100 %

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IM use : Reconstitute with 0.5% or 1% Lidocaine HCl Injection (without epinephrine).
IV use : Reconstitute with sterile water for injection.
Note :
1. After reconstitution carbon dioxide is formed. This may require venting.
2. If concurrent aminoglycosides or vancomycin therapy is indicated, administer each of these antibiotics separately.

Preparation of solutions for IM or IV bolus injection :
1. Drive the syringe needle through the vial closure and inject the recommended volume of diluents.
2. Withdraw the needle and shake the vial to give a clear solution.
3. Invert the vial. With the syringe piston fully depressed insert the needle into the solution. Withdraw the total volume of solution into the syringe ensuring that the needle remains in the solution. Small bubbles of carbon dioxide may be ignored.

Preparation of solutions of 1 g for IV infusion :
1. Drive the syringe needle through the vial closure and inject 10 mL diluent for 1 g vial.
2. Withdraw the needle and shake the vial to give a clear solution.
3. Insert a gas relief needle through the vial closure to relieve the internal pressure. Do not insert a gas relief needle until the product has dissolved.
4. Transfer the reconstituted solution to final delivery vehicle making up a total volume of at least 50 mL and administer by intravenous infusion over 15-30 minutes

Note :
To preserve product sterility, it is important that the gas relief needle is not inserted through the vial closure before the product has dissolved.
Solutions range from light yellow to amber depending on concentration, diluents and storage conditions used. Within the stated recommendations, product potency is not adversely affected by such colour variations.
Risk of contamination during preparation should be considered. Solution should be used immediately after preparation. The remaining solution should not be kept for further use.
Before use, check if there is any precipitation or colour change.
Should be infused via IV infusion equipment. It is not recommended to concomitantly **CEF-4** with other drug.

Administration :-
- **IM injection**
  Recommended IM injection sites are the upper outer quadrant of the gluteus maximus or lateral part of the thigh.
- **IV bolus injection**
  Ceftazidime should be injected over a period of 3-5 minutes.
- **IV infusion**
  Ceftazidime should be injected over a period of 15-30 minutes. The drug may be infused via IV infusion equipment.

OVERDOSAGE :-
Overdosage reactions have included seizure activity, encephalophathy, asterixis, neuromuscular excitability, and coma. Patients who receive an acute overdosage should be carefully observed and given supportive treatment. Serum level of Cef-4® can be reduced by hemodialysis or peritoneal dialysis.

INCOMPATIBILITIES :-
- It is not recommended to use sodium bicarbonate as a diluent.
- Ceftazidime and aminoglycosides should not be mixed in the same giving set or syringe. Precipitation has been reported when vancomycin mixed with Ceftazidime.

Therefore, when both drugs are to be administered by intermittent IV infusion, that they be given separately.

DRUG INTERACTIONS :-
Increased effect : Probenecid may decrease cephalosporin elimination; Aminoglycosides : *in vitro* studies indicate additive or synergistic effect against some strains of Enterobacteriaceae and *Pseudomonas aeruginosa*.
Increased toxicity : Furosemide, aminoglycosides may be a possible additive to nephrotoxicity.
Antagonism : Chloramphenicol has been shown to be antagonistic to beta-lactam antibiotics, including ceftazidime. This drug combination should be avoided when bactericidal activity is desired.

STORAGE :-
Store below 30°C. Protect from light.

SHELF LIFE :-
The expiry date is indicated on the packaging.
The shelf life after reconstitution and dilution when Cef-4® has been stored below:

Concentration (After Reconstitution)	Keep in Room Temperature (25°C)	Keep in Refrigerator (2-8°C)
Reconstituted suspension for IM (Approx. conc. 280 mg/mL)		
1% Lidocaine HCl Injection	12 hrs	12 hrs
Reconstituted solution for IV Bolus (Approx. conc. 100 mg/mL)		
Sterile water for injection (SWFI)	12 hrs	3 days
Diluted solution for IV Infusion (Approx. conc. 20 mg/mL)		
0.9% Sodium Chloride Injection (NSS)	12 hrs	3 days
5% Dextrose in water (D5W)	12 hrs	3 days

For IM: Recommend to 'use immediately' only
For IV: Recommend to 'use immediately' only or '3 days in a refrigerator'

Dosage form and packs :-
1 vial of dry substance equivalent to 500 mg or 1 g

Manufacturer :-
Siam Bheasach Co., Ltd.
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and 9 Soi Chokechairuammitr 3, Vibhavadi-Rangsit Road, Dindang, Dindang, Bangkok 10400, Thailand.

Product Registration Holder, Importer :-
SPG PHARMA (MALAYSIA) SDN BHD
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23 January 2024

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Actual Size 100 %

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